SESD-Assignment-4

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**Friday – E-Commerce Application Development**

**A software development team is building a complete E-Commerce Application involving multiple developers. The application includes user registration and login, product management, product search, shopping cart, order management, payment-related functionality, and an administrative dashboard. Multiple developers are expected to work simultaneously on common and separate components.**

**Assignment**

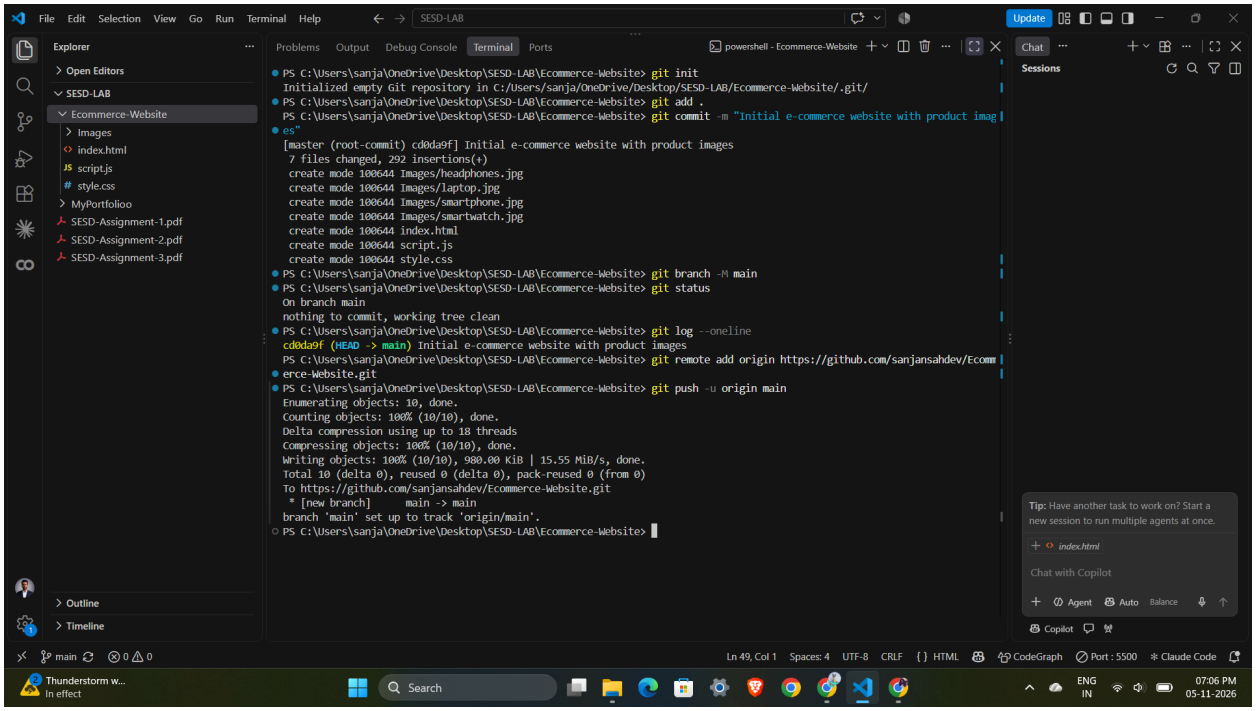
Develop the E-Commerce Application as a collaborative Git project using Git and a remote repository. Create the repository and design an appropriate branching and merging strategy for the development team. Create separate feature branches for the major application components and simulate multiple developers working concurrently. Developers should make multiple incremental changes, maintain meaningful commits, push and pull changes from the remote repository, and keep their branches synchronized with the main development line. Integrate the feature branches progressively and intentionally create complex merge-conflict situations by modifying common files and overlapping functionality from different branches. Analyze, resolve, and verify the conflicts without losing valid changes from any developer. Complete the final integration, verify the functionality and consistency of the complete application, and demonstrate the entire Git collaboration workflow using appropriate Git commands.

# E-Commerce Application Development

**Aim:** To develop and deploy a simple E-Commerce website using HTML, CSS, JavaScript, Git, and GitHub.

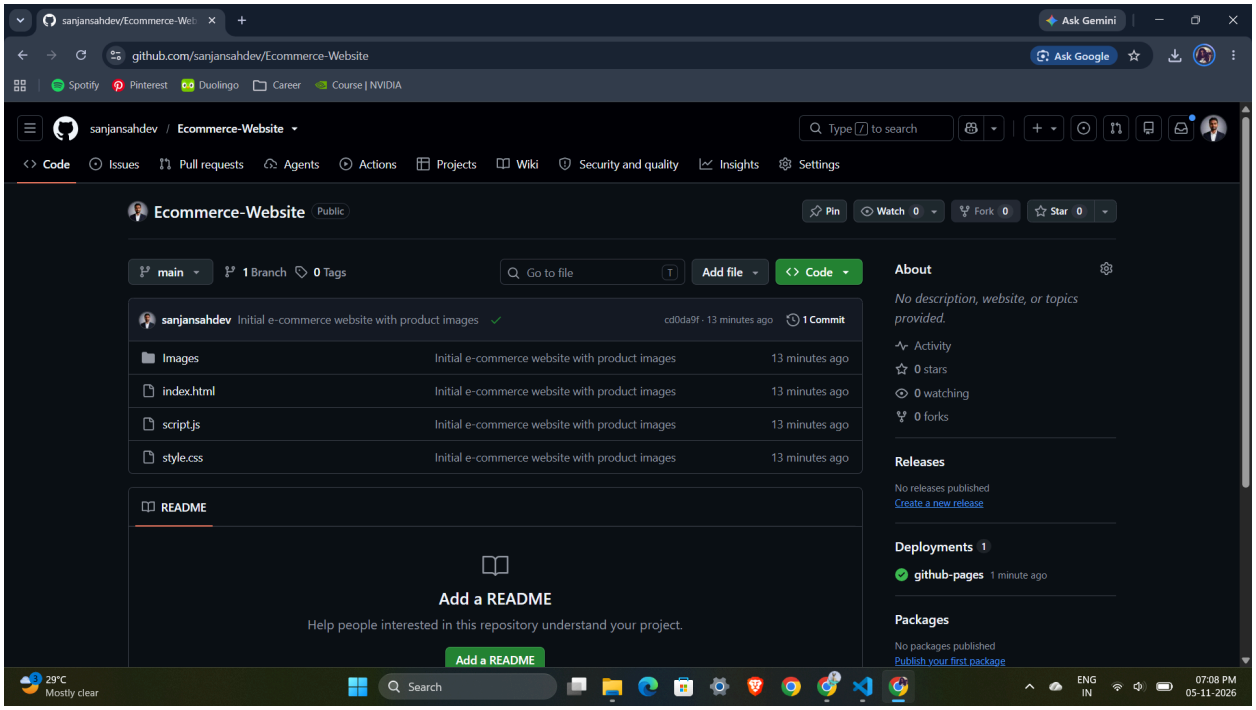
## 1. Git Initialization and GitHub Repository Creation

- Created an E-Commerce website using `index.html`, `style.css`, and `script.js`.
- Added product images to the project folder.
- Initialized the local Git repository using `git init`.
- Added and committed project files using `git add .` and `git commit`.
- Created a remote repository on GitHub.
- Uploaded the project using `git push -u origin main`.



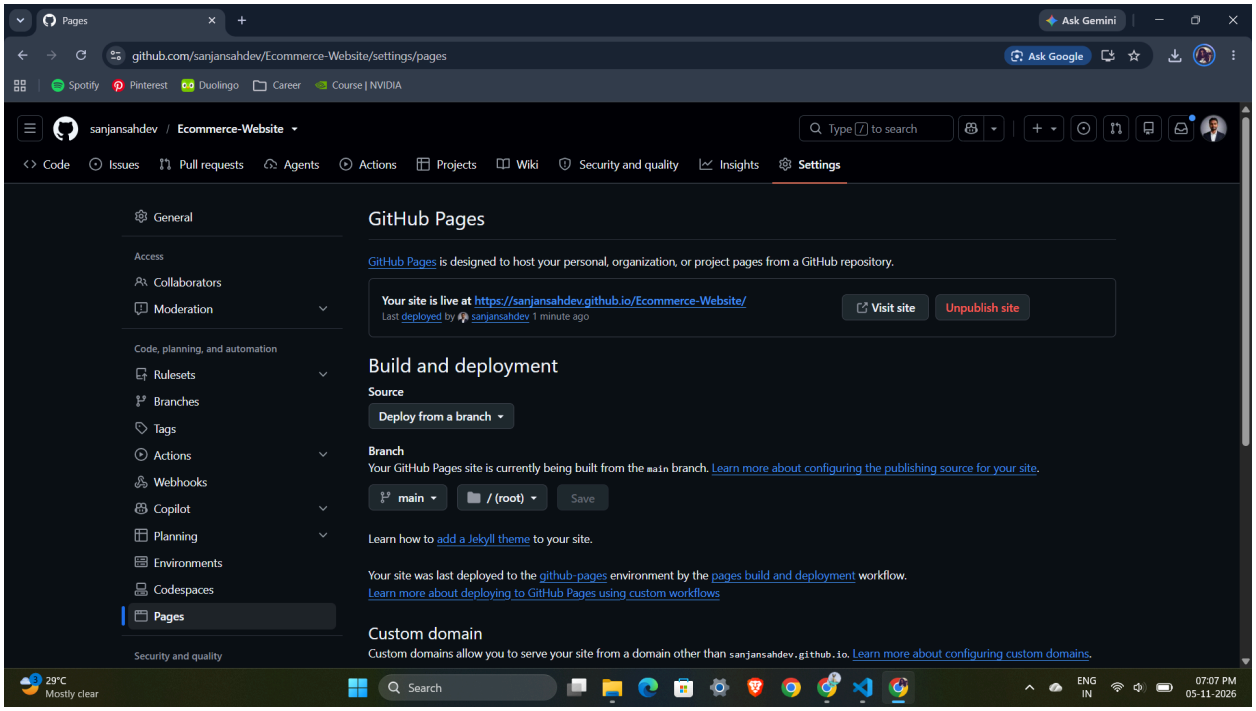

## 2. GitHub Pages Configuration and Deployment

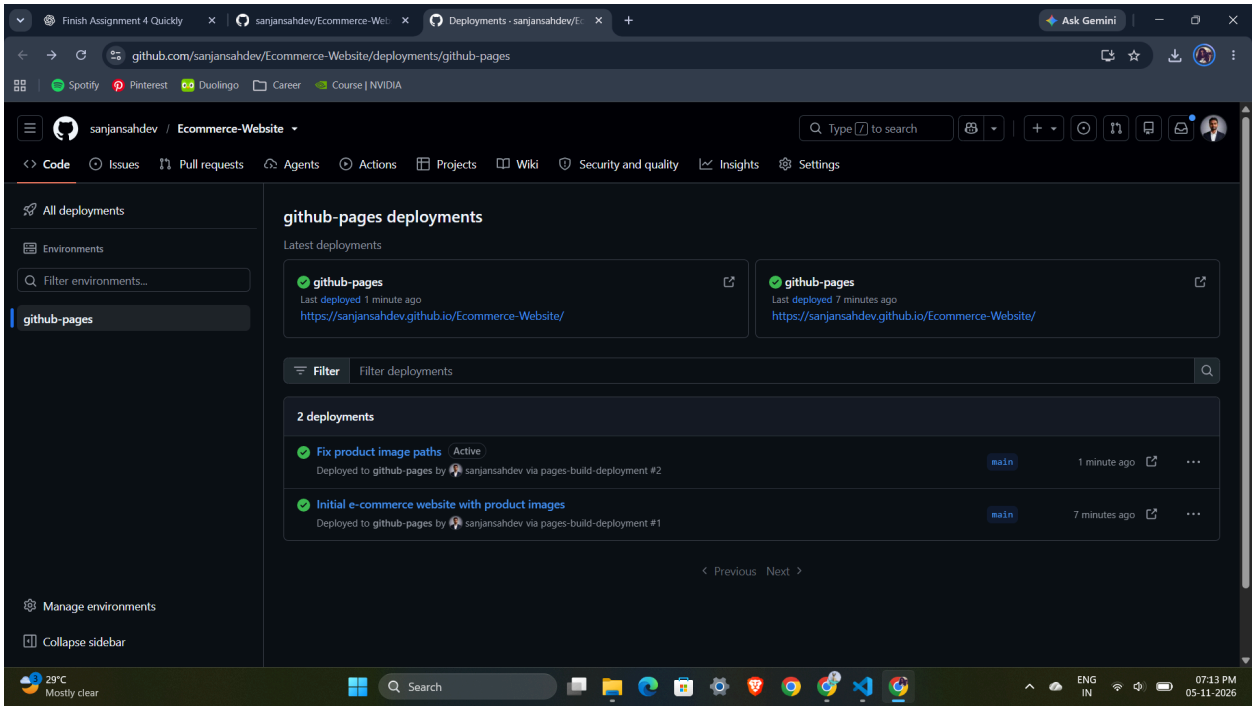
- Opened the GitHub repository settings.
- Navigated to the GitHub Pages section.
- Selected the `main` branch and `/ (root)` folder.
- Enabled GitHub Pages to host the website.




## 3. Deployment Verification

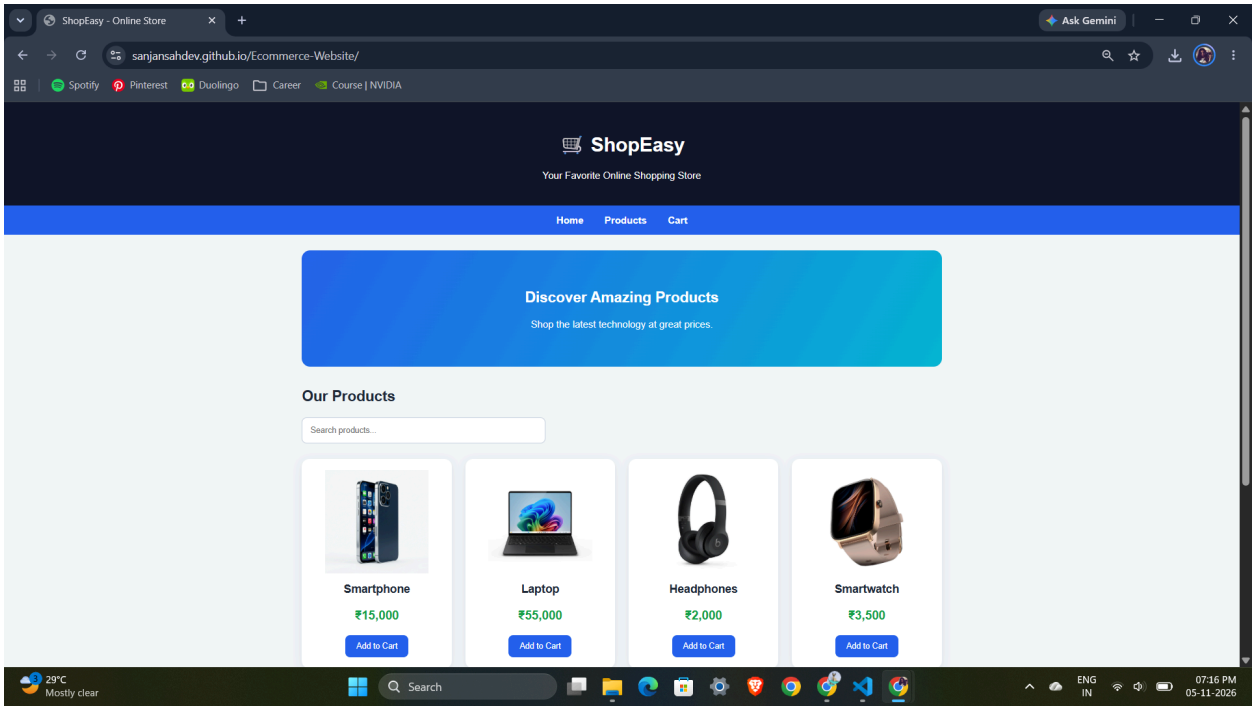
- Verified that the website was successfully deployed using GitHub Pages.
- Updated the product image paths and pushed the changes to GitHub.
- Confirmed the successful deployment of the updated version.






## 4. Final E-Commerce Website Output

- Opened the deployed E-Commerce website.
- Verified that product images, names, and prices were displayed correctly.
- Implemented product search, Add to Cart, and demo checkout functionality.
- Confirmed that the website was accessible through its public URL.




## Result

Successfully developed and deployed a simple E-Commerce website using HTML, CSS, JavaScript, Git, GitHub, and GitHub Pages.

**Live Website:** https://sanjansahdev.github.io/Ecommerce-Website/